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Ethical Hacking
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Ethical Hacking: Lab 1

```
mysql> create table works (employee_id int primary key, company_name varchar(20), salary decimal(10,2));
Query OK, 0 rows affected (0.07 sec)

mysql> insert into works values(1, 'TechCorp', 75000.00);
Query OK, 1 row affected (0.06 sec)

mysql> insert into works values(2, 'InnovateLtd', 62000.50);
Query OK, 1 row affected (0.00 sec)

mysql> insert into works values(3, 'HealthPlus', 54000.75);
Query OK, 1 row affected (0.00 sec)

mysql> select * from works;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
+-------------+--------------+----------+
3 rows in set (0.00 sec)

mysql> insert into works values(4, 'EduWorks', 48000.00);
Query OK, 1 row affected (0.01 sec)

mysql> insert into works values(5, 'GreenTech', 68000.00);
Query OK, 1 row affected (0.01 sec)

mysql> insert into works values(6, 'TechCorp', 80000.00);
Query OK, 1 row affected (0.00 sec)

mysql> insert into works values(7, 'InnovateLtd', 66000.20);
Query OK, 1 row affected (0.01 sec)

mysql> insert into works values(8, 'HealthPlus', 50000.10);
Query OK, 1 row affected (0.05 sec)

mysql> insert into works values(9, 'EduWorks', 51000.00);
Query OK, 1 row affected (0.06 sec)
```

```
mysql> insert into works values(10, 'GreenTech', 72000.00);
Query OK, 1 row affected (0.05 sec)

mysql> select * from works;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
|           4 | EduWorks     | 48000.00 |
|           5 | GreenTech    | 68000.00 |
|           6 | TechCorp     | 80000.00 |
|           7 | InnovateLtd  | 66000.20 |
|           8 | HealthPlus   | 50000.10 |
|           9 | EduWorks     | 51000.00 |
|          10 | GreenTech    | 72000.00 |
+-------------+--------------+----------+
10 rows in set (0.00 sec)
```

FIGURE 1. Create Table "Works"

```
mysql> select distinct(company_name) from works;
+--------------+
| company_name |
+--------------+
| TechCorp     |
| InnovateLtd  |
| HealthPlus   |
| EduWorks     |
| GreenTech    |
+--------------+
5 rows in set (0.00 sec)
```

FIGURE 2. Distinct Company Names

```
mysql> select * from works;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
|           4 | EduWorks     | 48000.00 |
|           5 | GreenTech    | 68000.00 |
|           6 | TechCorp     | 80000.00 |
|           7 | InnovateLtd  | 66000.20 |
|           8 | HealthPlus   | 50000.10 |
|           9 | EduWorks     | 51000.00 |
|          10 | GreenTech    | 72000.00 |
+-------------+--------------+----------+
10 rows in set (0.00 sec)
```

FIGURE 3. All Records

```
mysql> select * from works;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
|           4 | EduWorks     | 48000.00 |
|           5 | GreenTech    | 68000.00 |
|           6 | TechCorp     | 80000.00 |
|           7 | InnovateLtd  | 66000.20 |
|           8 | HealthPlus   | 50000.10 |
|           9 | EduWorks     | 51000.00 |
|          10 | GreenTech    | 72000.00 |
+-------------+--------------+----------+
10 rows in set (0.00 sec)
```

FIGURE 4. Display employeeid companyname salary

```
mysql> select * from works where salary>60000;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           2 | InnovateLtd  | 62000.50 |
|           5 | GreenTech    | 68000.00 |
|           6 | TechCorp     | 80000.00 |
|           7 | InnovateLtd  | 66000.20 |
|          10 | GreenTech    | 72000.00 |
+-------------+--------------+----------+
6 rows in set (0.00 sec)
```

FIGURE 5. Salary more than 60,000

```
mysql> select * from works where company_name='TechCorp';
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           6 | TechCorp     | 80000.00 |
+-------------+--------------+----------+
2 rows in set (0.00 sec)
```

FIGURE 6. Employess working in TechCorp

```
mysql> select * from works where salary>=50000&&salary<=70000;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
|           5 | GreenTech    | 68000.00 |
|           7 | InnovateLtd  | 66000.20 |
|           8 | HealthPlus   | 50000.10 |
|           9 | EduWorks     | 51000.00 |
+-------------+--------------+----------+
6 rows in set (0.00 sec)
```

FIGURE 7. Salary between 50000 and 70000

```
mysql> select max(salary) from works;
+--------------+
| max(salary) |
+--------------+
|    80000.00 |
+--------------+
1 row in set (0.00 sec)
```

FIGURE 8. Maximum Salary

```
mysql> select min(salary) from works;
+--------------+
| min(salary) |
+--------------+
|    48000.00 |
+--------------+
1 row in set (0.00 sec)
```

FIGURE 9. Minimum Salary

```
mysql> select avg(salary) from works;
+--------------+
| avg(salary)  |
+--------------+
| 62600.155000 |
+--------------+
1 row in set (0.00 sec)
```

FIGURE 10. Average salary of all employees

```
mysql> select company_name,sum(salary) as total_salary from works group by company_name;
+--------------+--------------+
| company_name | total_salary |
+--------------+--------------+
| EduWorks     |     99000.00 |
| GreenTech    |    154000.00 |
| HealthPlus   |    104000.85 |
| InnovateLtd  |    128000.70 |
| TechCorp     |    155000.00 |
+--------------+--------------+
5 rows in set (0.00 sec)
```

FIGURE 11. Total salary paid by companies

```
mysql> select count(*) as total_employees from works;
+----------------+
| total_employees |
+----------------+
|             10 |
+----------------+
1 row in set (0.00 sec)
```

FIGURE 12. Total no. of employees

```
mysql> select company_name,avg(salary) from works group by company_name;
+--------------+--------------+
| company_name | avg(salary)  |
+--------------+--------------+
| EduWorks     | 49500.000000 |
| GreenTech    | 70000.000000 |
| HealthPlus   | 52000.425000 |
| InnovateLtd  | 64000.350000 |
| TechCorp     | 77500.000000 |
+--------------+--------------+
5 rows in set (0.00 sec)
```

FIGURE 13. Average salary of each company

```
mysql> select company_name,count(*) as num_employees from works group by company_name;
+--------------+---------------+
| company_name | num_employees |
+--------------+---------------+
| EduWorks     |             2 |
| GreenTech    |             2 |
| HealthPlus   |             2 |
| InnovateLtd  |             2 |
| TechCorp     |             2 |
+--------------+---------------+
5 rows in set (0.01 sec)
```

FIGURE 14. No. of employees in each company

```
mysql> select company_name,avg(salary) as avg_salary from works group by company_name having
  avg_salary>65000;
+--------------+--------------+
| company_name | avg_salary   |
+--------------+--------------+
| GreenTech    | 70000.000000 |
| TechCorp     | 77500.000000 |
+--------------+--------------+
2 rows in set (0.00 sec)
```

FIGURE 15. Average salary more than 65000

```
mysql> select company_name,count(*) as num_employees from works group by company_name having
 num_employees>1;
+--------------+---------------+
| company_name | num_employees |
+--------------+---------------+
| EduWorks     |             2 |
| GreenTech    |             2 |
| HealthPlus   |             2 |
| InnovateLtd  |             2 |
| TechCorp     |             2 |
+--------------+---------------+
5 rows in set (0.00 sec)
```

FIGURE 16. Companies with more than 1 employee

```
mysql> select * from works order by salary desc;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           6 | TechCorp     | 80000.00 |
|           1 | TechCorp     | 75000.00 |
|          10 | GreenTech    | 72000.00 |
|           5 | GreenTech    | 68000.00 |
|           7 | InnovateLtd  | 66000.20 |
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
|           9 | EduWorks     | 51000.00 |
|           8 | HealthPlus   | 50000.10 |
|           4 | EduWorks     | 48000.00 |
+-------------+--------------+----------+
10 rows in set (0.00 sec)
```

FIGURE 17. Employees with salary in descending order

```
mysql> select * from works order by company_name,salary;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           4 | EduWorks     | 48000.00 |
|           9 | EduWorks     | 51000.00 |
|           5 | GreenTech    | 68000.00 |
|          10 | GreenTech    | 72000.00 |
|           8 | HealthPlus   | 50000.10 |
|           3 | HealthPlus   | 54000.75 |
|           2 | InnovateLtd  | 62000.50 |
|           7 | InnovateLtd  | 66000.20 |
|           1 | TechCorp     | 75000.00 |
|           6 | TechCorp     | 80000.00 |
+-------------+--------------+----------+
10 rows in set (0.00 sec)
```

FIGURE 18. Sorted by company name and salary

```
mysql> select * from works order by salary desc limit 3;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           6 | TechCorp     | 80000.00 |
|           1 | TechCorp     | 75000.00 |
|          10 | GreenTech    | 72000.00 |
+-------------+--------------+----------+
3 rows in set (0.00 sec)
```

FIGURE 19. Highest 3 salaries

```
mysql> select * from works where salary>(select avg(salary) from works);
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           5 | GreenTech    | 68000.00 |
|           6 | TechCorp     | 80000.00 |
|           7 | InnovateLtd  | 66000.20 |
|          10 | GreenTech    | 72000.00 |
+-------------+--------------+----------+
5 rows in set (0.00 sec)
```

FIGURE 20. Salary more than average salary

```
mysql> select * from works where salary=(select max(salary) from works);
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           6 | TechCorp     | 80000.00 |
+-------------+--------------+----------+
1 row in set (0.00 sec)
```

FIGURE 21. Employees with highest salary

```
mysql> select * from works where salary<(select avg(salary) from works group by company_name
 having company_name='TechCorp');
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
|           4 | EduWorks     | 48000.00 |
|           5 | GreenTech    | 68000.00 |
|           7 | InnovateLtd  | 66000.20 |
|           8 | HealthPlus   | 50000.10 |
|           9 | EduWorks     | 51000.00 |
|          10 | GreenTech    | 72000.00 |
+-------------+--------------+----------+
9 rows in set (0.00 sec)
```

FIGURE 22. Less than average salary of TechCorp

```
mysql> update works set salary=salary+salary*0.1 where company_name='GreenTech';
Query OK, 2 rows affected (0.01 sec)

mysql> select * from works;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
|           4 | EduWorks     | 48000.00 |
|           5 | GreenTech    | 74800.00 |
|           6 | TechCorp     | 80000.00 |
|           7 | InnovateLtd  | 66000.20 |
|           8 | HealthPlus   | 50000.10 |
|           9 | EduWorks     | 51000.00 |
|          10 | GreenTech    | 79200.00 |
+-------------+--------------+----------+
10 rows in set (0.01 sec)
```

FIGURE 23. Salary of GreenTech employees increased by 10

```
mysql> select distinct(company_name) from works;
+--------------+
| company_name |
+--------------+
| TechCorp     |
| InnovateLtd  |
| HealthPlus   |
| EduWorks     |
| GreenTech    |
+--------------+
5 rows in set (0.00 sec)
```

FIGURE 24. Distinct company names

```
mysql> delete from works where salary<50000;
Query OK, 1 row affected (0.01 sec)

mysql> select * from works;
+-------------+--------------+----------+
| employee_id | company_name | salary   |
+-------------+--------------+----------+
|           1 | TechCorp     | 75000.00 |
|           2 | InnovateLtd  | 62000.50 |
|           3 | HealthPlus   | 54000.75 |
|           5 | GreenTech    | 74800.00 |
|           6 | TechCorp     | 80000.00 |
|           7 | InnovateLtd  | 66000.20 |
|           8 | HealthPlus   | 50000.10 |
|           9 | EduWorks     | 51000.00 |
|          10 | GreenTech    | 79200.00 |
+-------------+--------------+----------+
9 rows in set (0.00 sec)
```

FIGURE 25. Delete salary less than 50000